3D Printing and Modeling

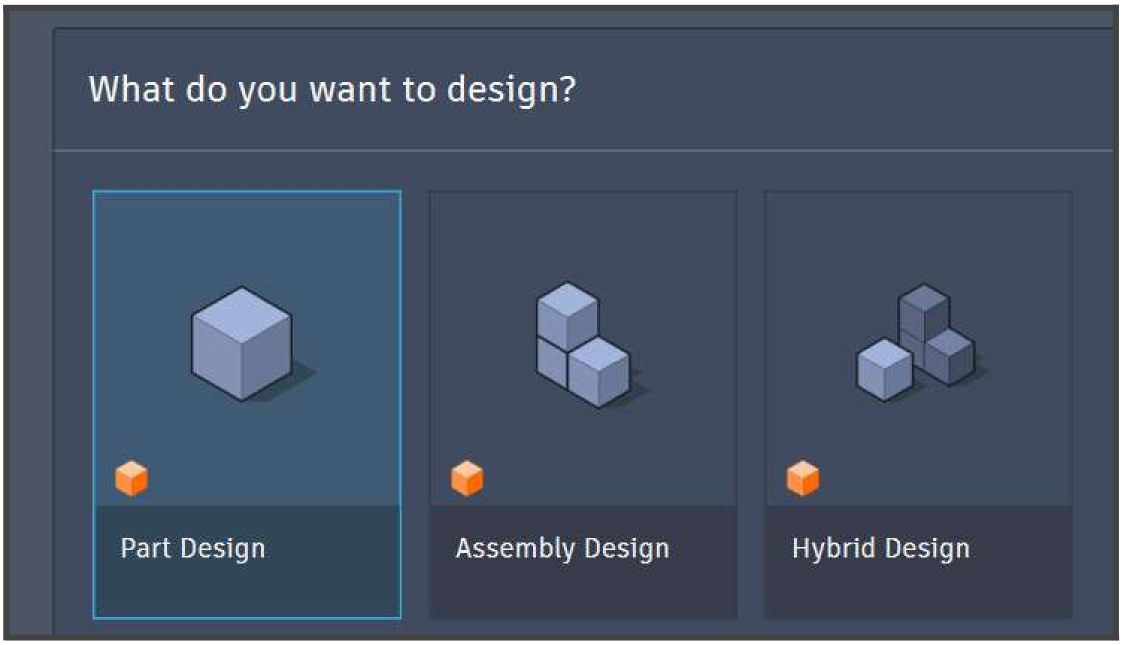
5th lab: Assembly Design

0. Resources

1. Autodesk - Practice Exercise - Project Sample (7-8 min)
2. Fusion 360 - Joints Practice - All Joints Visualized (13 min)
3. Creating MOVING HINGE JOINTS - Door Hinge Example (7-8 min)
4. Creating a Hinge with Fusion 360 - Complete Door Hinge (37min)
5. Pin-Slot Joint in Fusion 360 - Pin-Slot Joint Tutorial (3 min)

1. What is Assembly Design?

Up until now, all the designs we've created had been done in one single file of either **Part** or **Hybrid Design** type. It's time we discuss what they mean and how working with multiple files is done in fusion.



Starting in January 2026, Autodesk Fusion introduced a system called **Intent-Driven Design**, which adds two new working modes: **Part Design** and **Assembly Design** alongside the classic approach, now renamed **Hybrid Design**.

Hybrid design is similar to what we've done so far: creating multiple bodies and components in the same file and placing them around as we wish. The new approach wishes to separate the design stage from the assembly stage by using separate files. It relies on creating multiple components, each in its own file, then importing them into a separate **Assembly File** where they are joined together. Changes made to the original **Design file** will also appear in the final assembly.

In this laboratory, you will see the words **Body** and **Component** used multiple times. You've met them before, but the difference between them was not mentioned much. So here it is:
Bodies: A body is any 3D shape that we created while extruding, revolving etc. When we cut an object in half, we get 2 separate bodies. The bodies are the objects we export in STL for 3D printing.

Components: Components are containers (like folders) that include bodies, sketches, planes and more. Each component comes with its own timeline. Evidently, components are more powerful than bodies and they allow us to assemble them as we wish.

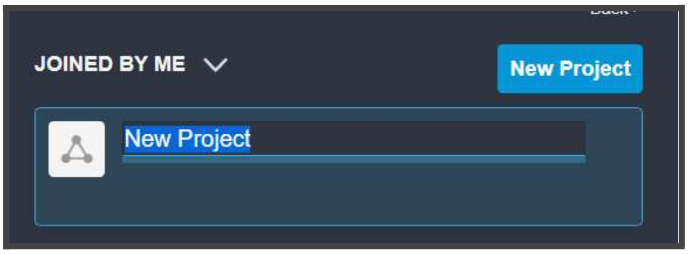
1.1 Working on our project

We've discussed creating objects from scratch for the last four laboratories. This time we will skip the design stage and focus on assembling the components.

First, we will need to create a new project. It will be the home for all of our components files and our assembly files.

Go to the **Data Panel** and click **New Project**. Give it any name you like.

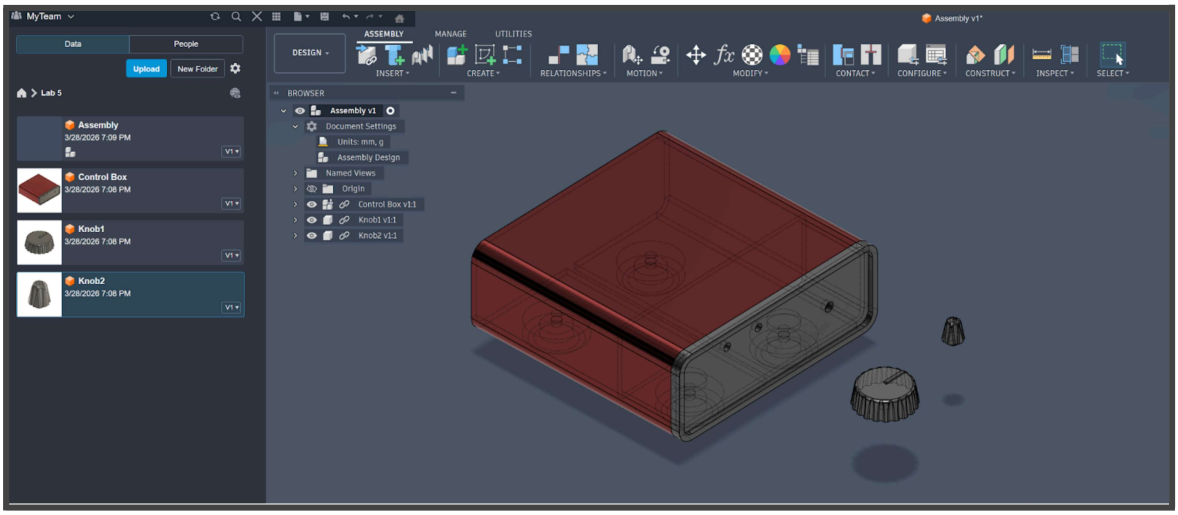
Now download the archive from Teams and import all files from the **Practice Exercise** folder into your project.



Your Project > **Upload** > **Drag the 3 files**.

Next, create a new Assembly File and save (**Ctrl + S**) it into the project. You will now have 4 files in your project. With the assembly open, just drag the components onto the screen. You will be prompted to position each one after dragging it. Place them where you can see each one clearly.

In the end you should see something like this:



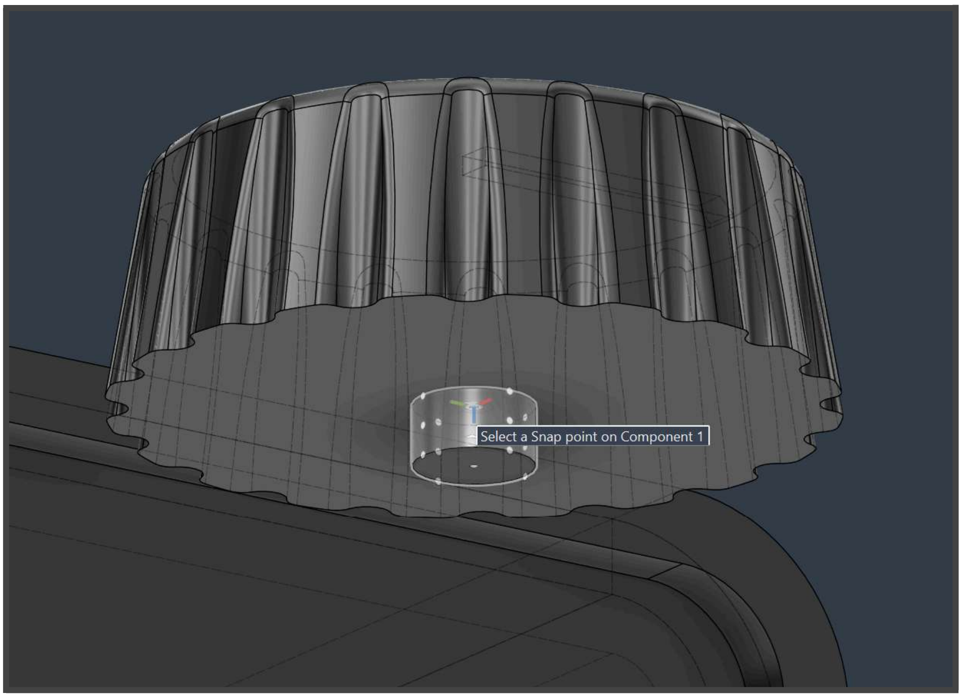
As you can see, the toolbar in the assembly file is very different from the one we are used to. The tools in here are meant to help us join and connect components.

Components used in this exercise are part of Autodesk's Training Exercises found here: https://www.autodesk.com/learn/ondemand/tutorial/introduction-to-fusion-assembly-modeling

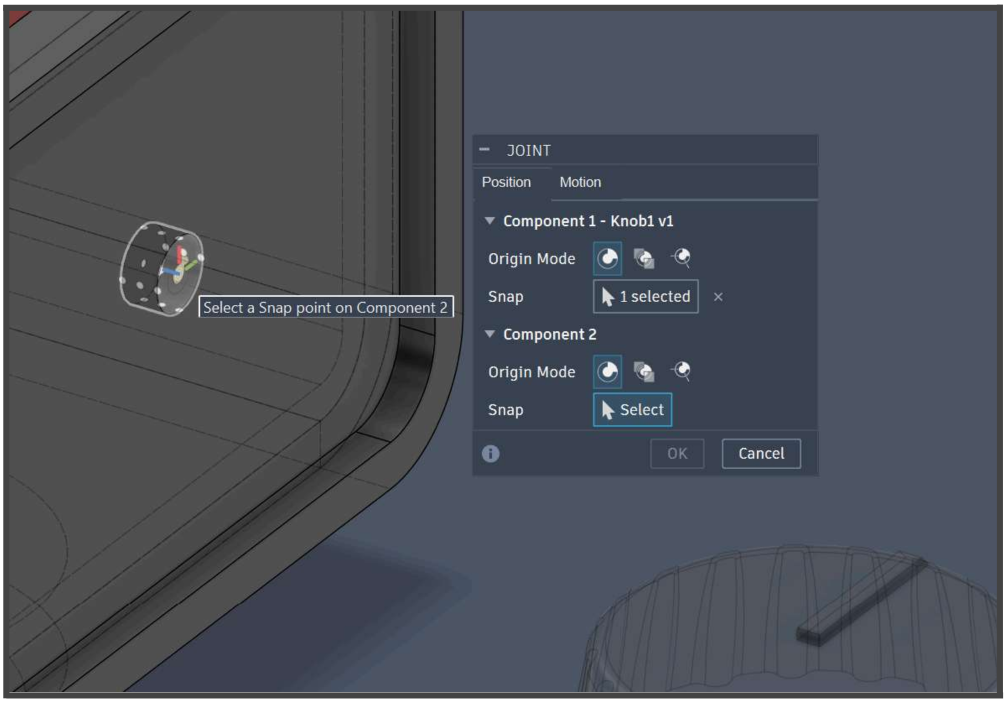
1.2 Joints

Joints are the way Fusion 360 defines the position and allowed movement between two components in an assembly.

Go to **Relationships** > **Joint**. You will then be prompted to select the two components you wish to join and a respective **Snap Point** for each one. Snap points are the points that will touch as the joint is formed. Select the first point to be inside the knob's stem. This will also act as its axis of rotation.

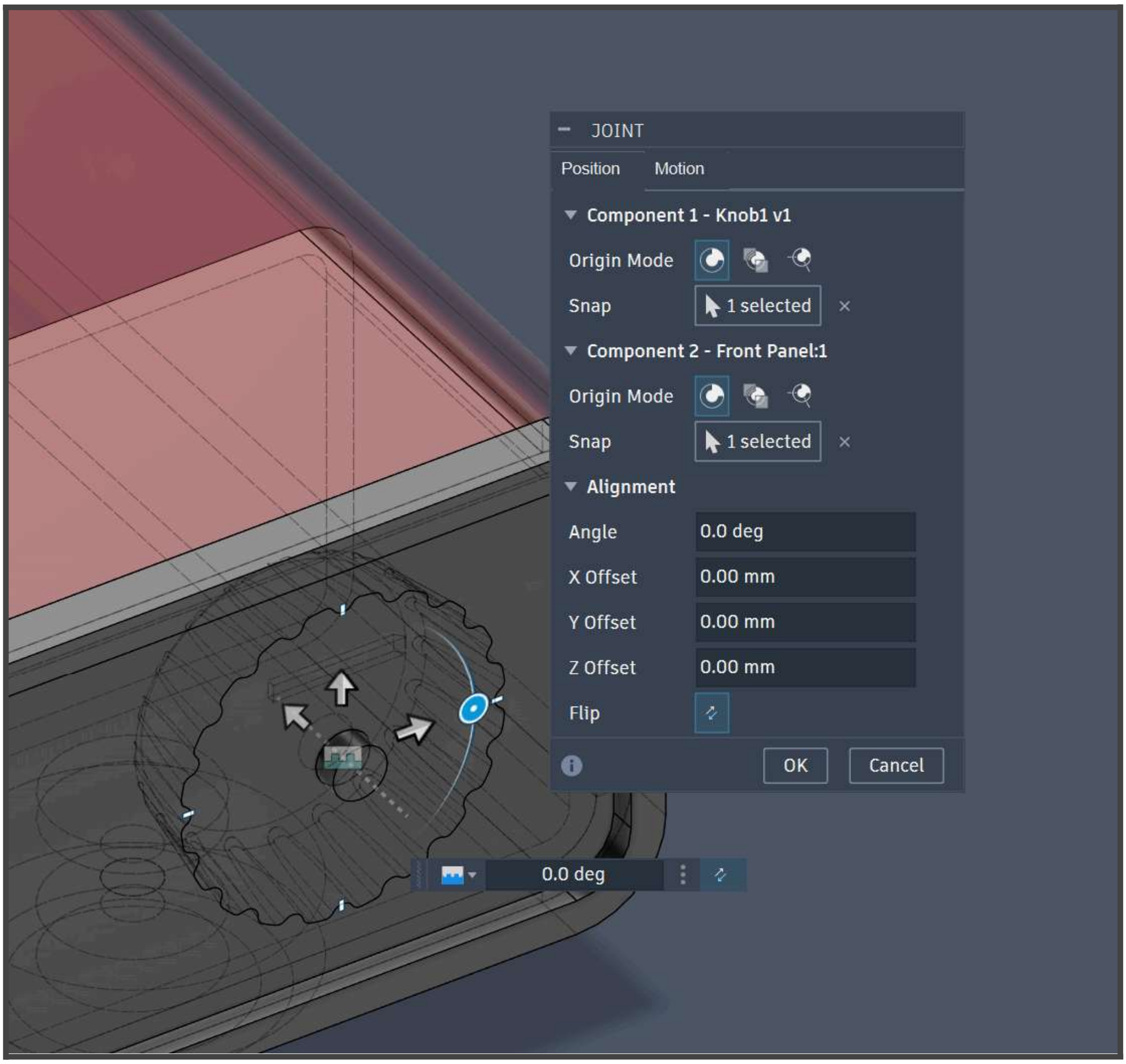


The second point should be on the housing:



As you can see, Fusion automatically generates snap points on each component. They are not always on the perimeter of the object and not even part of the object itself.

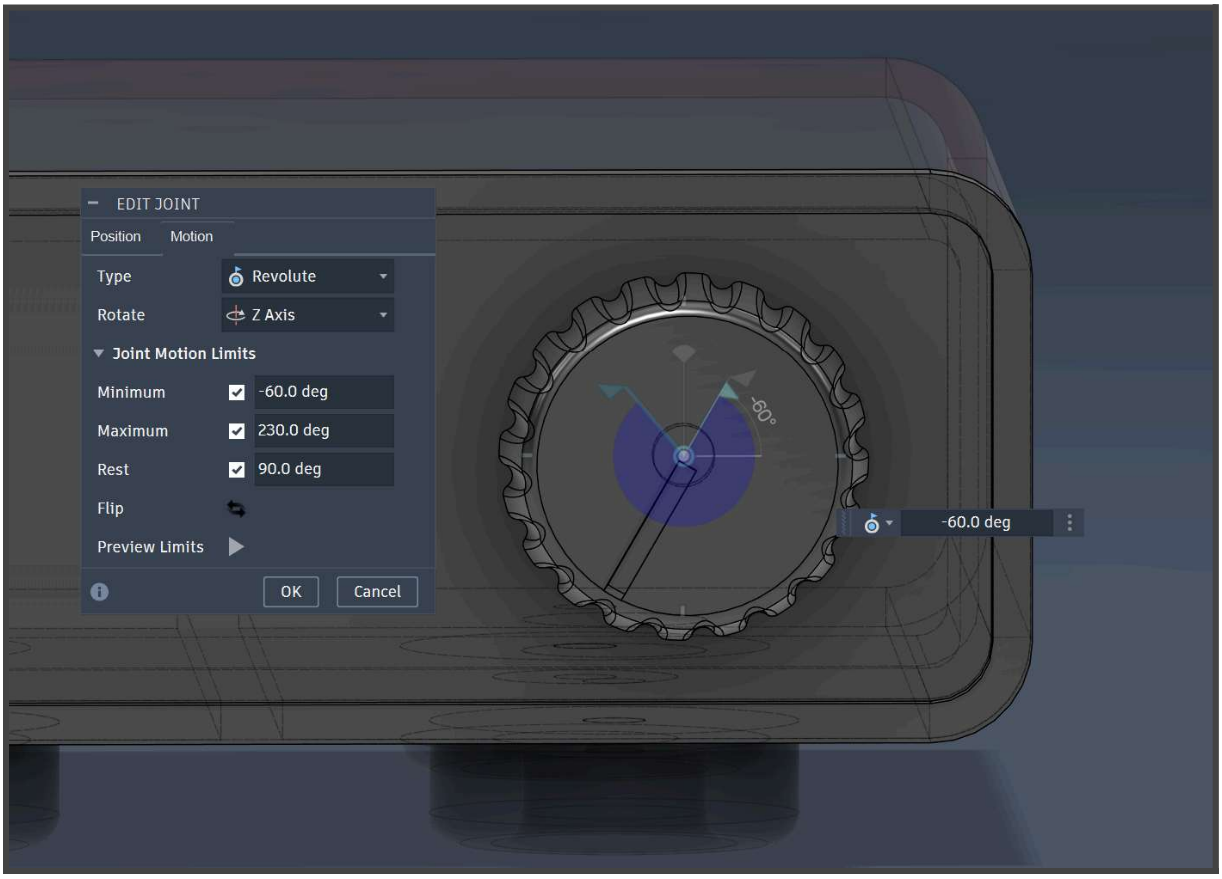
After selecting the second snap point, one of the objects will move into position to connect with the other. The created joint will likely not be right the first time, so we will need to tweak the settings to get it in position. Firstly, if the knob is inside the housing (see picture below) click on **Flip** to reverse it.



Next we will need to select the type of joint we wish to create. The main types are:

- **Rigid**: Objects are fixed to each other. Like glueing or friction fitting in the real world.
- **Revolute**: One object rotates around a fixed axis.
- **Slider:** One object slides on the surface of another.
- **Pin-Slot:** Similar to revolute, but the axis or rotation is not fixed.
- **Ball**: Allows the object to rotate in all directions around a single fixed point.

I recommend you go through multiple of them and click **Preview Motion** to get an idea of how each one works. For our knob we will be using **Revolve.** You can also adjust a minimum, maximum and rest (default) position. This helps us limit the rotation of the knob.



Click ok. A little blue arrow will appear, which you can double click to play with the joint.

1.3 Exercise 0:

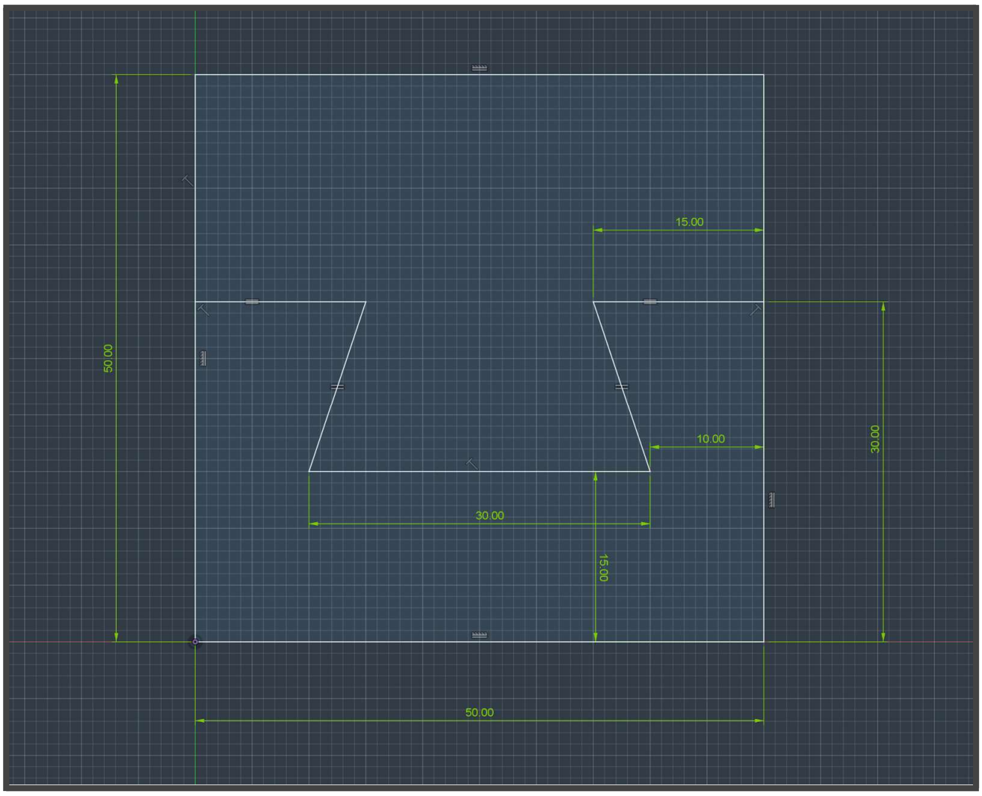
Add working knobs to the other two slots.

> You can also create a Joint by positioning the knob in the correct position on the housing (using **Move**) and selecting **Relationships** > **As-Built Joint (Shift + J).** This saves time by not having to find the correct Snap Point.

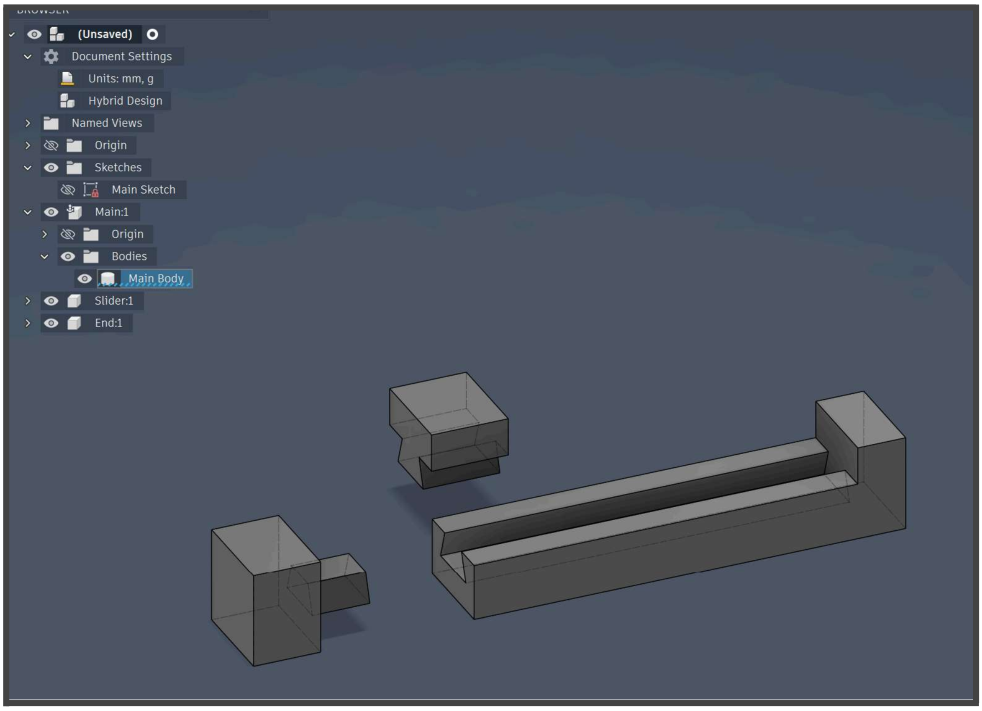
2. Exercises

Exercise 1:

Create a new file inside the project. It can be a simple hybrid file. Remember, hybrid files allow us to design and assemble components in the same file.

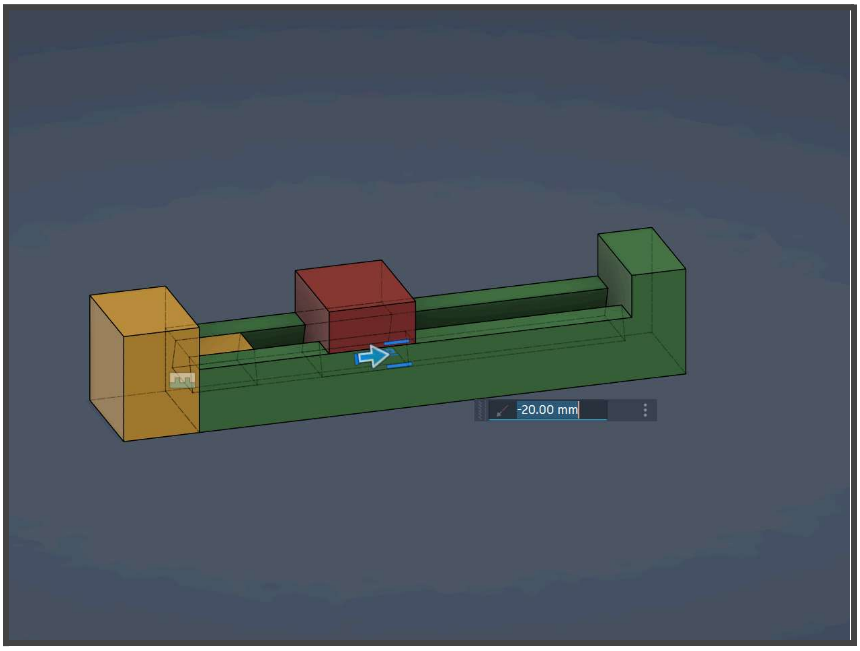


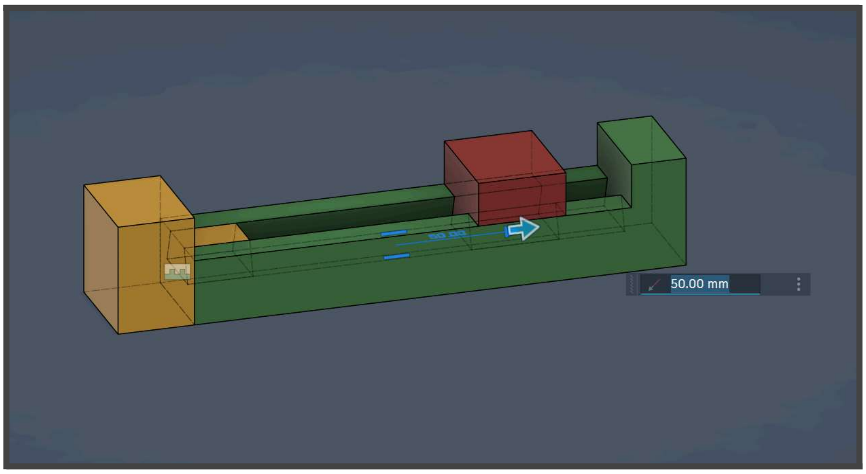
Extrude the following components from the sketch. Exact dimensions are not important.



Make sure to create **new components**, not new bodies inside the same component, since only components can be jointed. You can select **New Component** during extrude, or right click an existing body to turn it into a component.

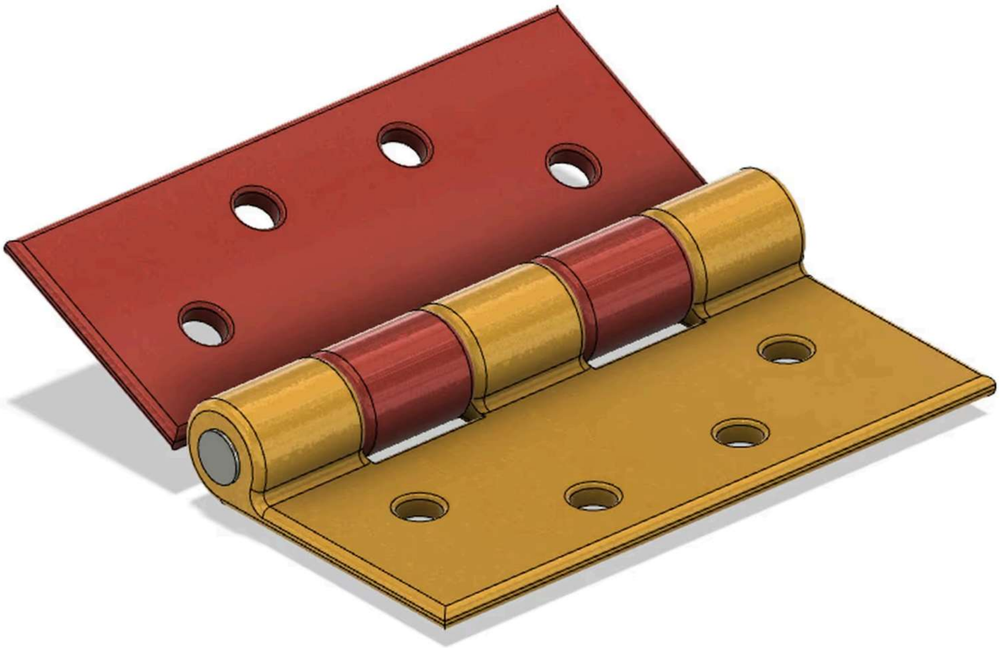
Fix the ends of the rail to each other using a **rigid joint**. Then take the middle piece and make it slide inside the railing. Assemble them as seen below and make them move. Make sure the minimum and maximum are set correctly so that the object doesn't slide outside the railing.

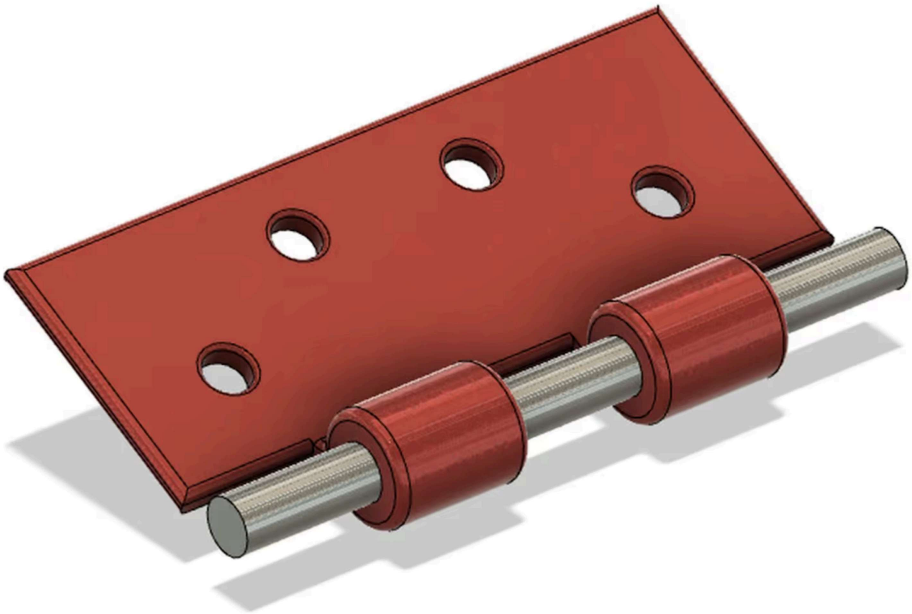




Exercise 2:

Recreate this door hinge





Credits: ***dwinanto*** (https://www.cadcrowd.com/3d-models/hinge-1)